

Juvenile Recidivism Predictor Study: Project Brief

Sep 26, 2026 · @Travis

Tools and skills

Tools: Python (pandas, NumPy, statsmodels, scikit-learn, Seaborn, Matplotlib) and R.

Skills: logistic regression and odds ratios, controlling for confounders, interaction terms and likelihood-ratio tests, statistical power, bootstrap confidence intervals, train/test evaluation, AUC, calibration, and subgroup fairness checks.skills

Purpose

I built a full analysis pipeline to test which interventions are associated with lower re-offending among justice-involved male adolescents, and how that answer changes across risk profiles. The project compares Cognitive Behavioral Therapy (CBT), vocational training, and "tough love" programs against standard supervision, with re-offending measured at 12 and 24 months.

The data is synthetic: 3,000 simulated youth records. Real juvenile records require a data use agreement and IRB approval. Because I set the true effects when generating the data, I could check whether each method recovered them before applying it to real records.

Approach

- **Data design.** Each record has age, offender status, school attendance, family support, a composite risk score, a program, and re-offense outcomes. Program placement is not random: higher-risk youth were steered toward tough love, mirroring how placements happen in practice.
- **Logistic regression.** I compared a naive model (program only) with an adjusted model controlling for prior record, risk, and age.
- **Interactions.** I tested whether program effects differ by offender status and by risk tier, chosen in advance from the risk principle and deterrence research.
- **Visualization.** A four-panel Seaborn figure: raw rates, a forest plot of odds ratios, and two interaction plots with bootstrap confidence intervals.

- **Risk prediction.** A scikit-learn model using intake data only, graded on held-out youth. It reports probabilities with no cutoff and was evaluated on calibration.
- **Replication.** I reran the regressions in R and got identical results.

Key findings

After adjustment, CBT and vocational training were associated with lower re-offending, and tough love showed no significant effect.

Program vs. standard supervision	Naive odds ratio	Adjusted odds ratio (95% CI)
CBT	0.38	0.35 (0.28 to 0.45)
Vocational training	0.53	0.59 (0.46 to 0.76)
Tough love	1.38	1.12 (0.89 to 1.39)

- **Confounding.** Tough love looked harmful in the raw data ($p = .002$). Much of that came from which youth were placed there. After controls, $p = .33$.
- **Statistical power.** Both interaction effects exist in the data, but neither was significant with 3,000 youth. At 9,000, the risk tier effect became significant ($p = .02$). The smaller offender status effect did not ($p = .24$).
- **Risk principle.** CBT's benefit was largest for high-risk youth: about 23 points lower predicted 24-month recidivism than standard supervision, versus about 13 points for low-risk youth.
- **Prediction.** The intake model reached an AUC of 0.69. Its probabilities matched actual rates for most youth and for both offender groups. It overestimated the highest-risk group by about 7 points.

Limitations

- **Data bias.** I set the effects, so the project shows that the methods work. It cannot show that any program works for real youth. The data has no race, ethnicity, or neighborhood variables, so it cannot test the uneven error rates found in real risk tools.
- **Reporting gaps.** Real recidivism is measured through re-arrest, which also reflects how heavily an area is policed. The simulation has no missing data, no loss to follow-up, and no record of program quality or completion.

- **What it can't tell you.** An observational design cannot establish cause. A 0.69 AUC means many individual predictions will be wrong.